Finch beaks and the need for statistics

STATISTICAL THINKING IN PYTHON (PART 2)

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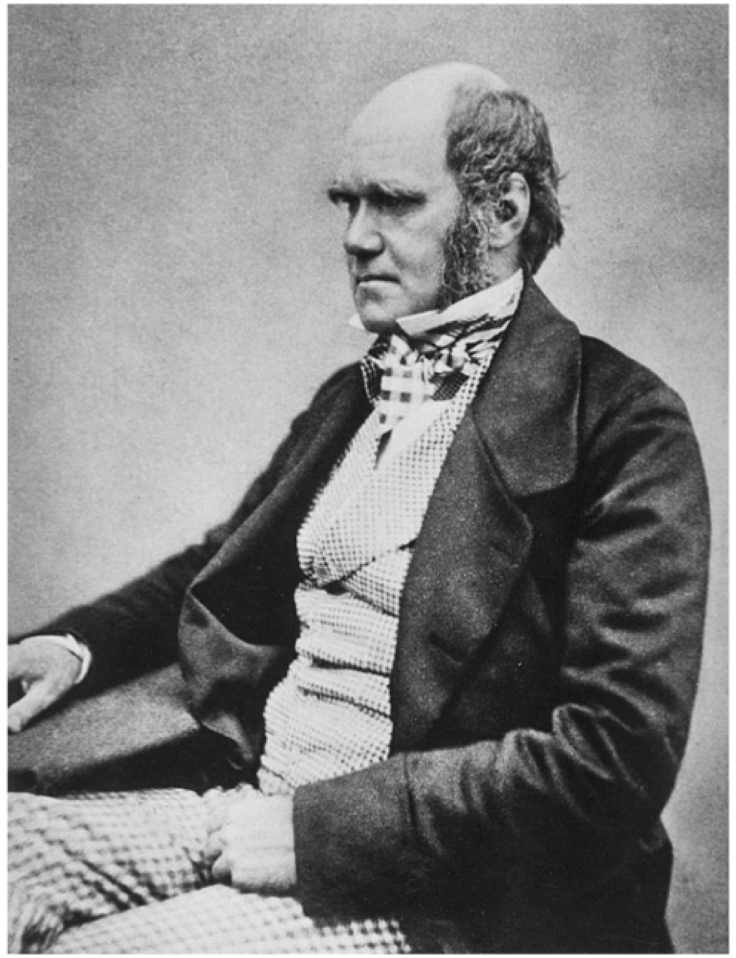
Your well-equipped toolbox

- Graphical and quantitative EDA
- Parameter estimation
- Confidence interval calculation
- Hypothesis testing



[1] Image: Public domain, US

[1] Image: NASA

The island of Daphne Major

[1] Image: Grant and Grant, 2014

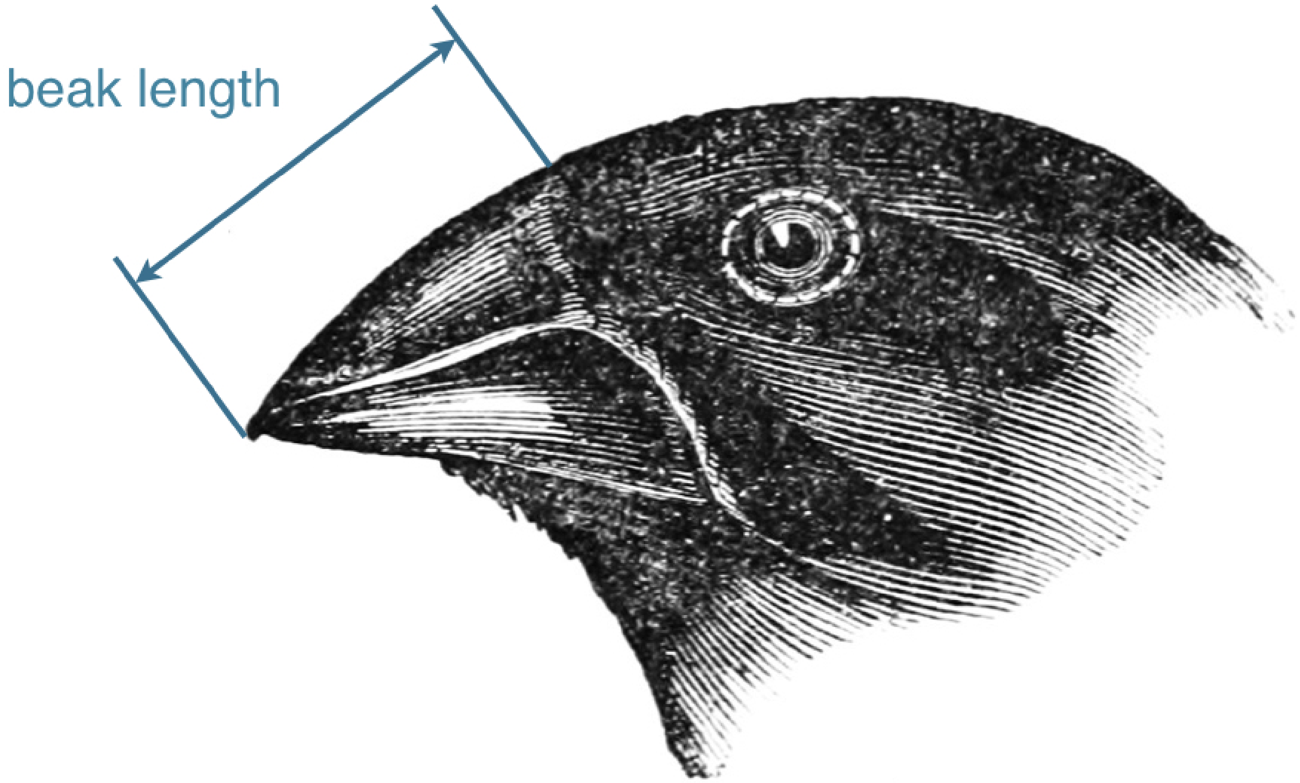
The finches of Daphne Major

[1] Source: John Gould, public domain

Our data source

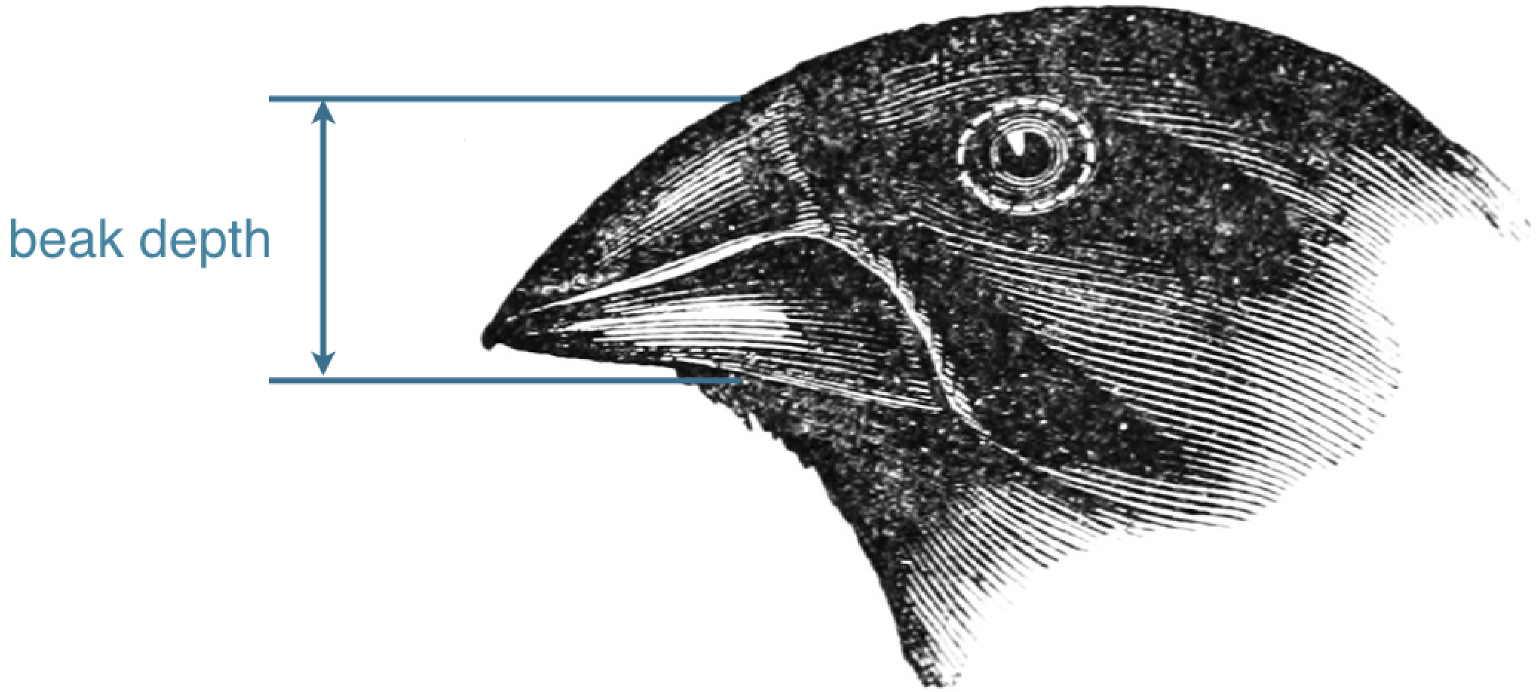
- Peter and Rosemary Grant
  - 40 Years of Evolution: Darwin's Finches on Daphne Major Island
  - Princeton University Press, 2014
- Data acquired from Dryad Digital Repository
  - http://dx.doi.org/10.5061/dryad.g6g3h

The dimensions of the finch beak



[1] Source: John Gould, public domain

The dimensions of the finch beak



[1] Source: John Gould, public domain

Investigation of G. scandens beak depth

- EDA of beak depths in 1975 and 2012
- Parameter estimates of mean beak depth
- Hypothesis test: did the beaks get deeper?

Let's practice!

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Variation in beak shapes

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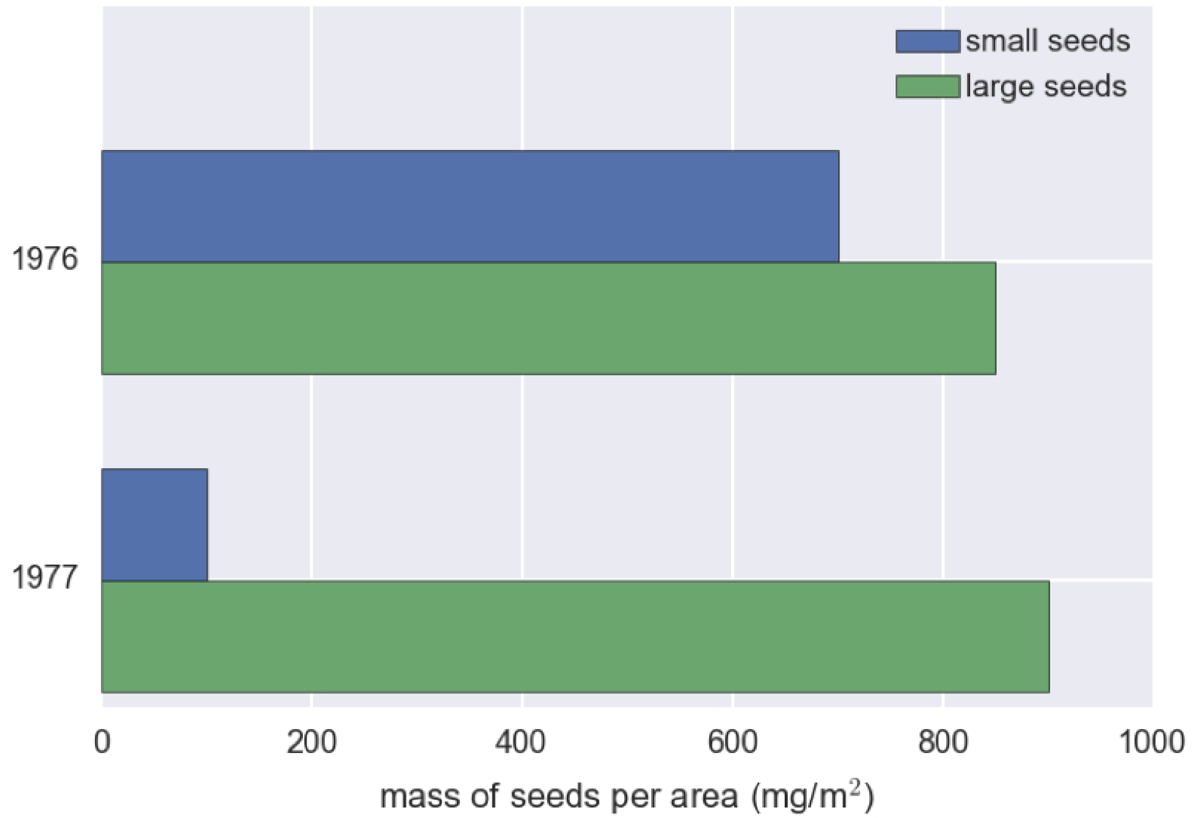
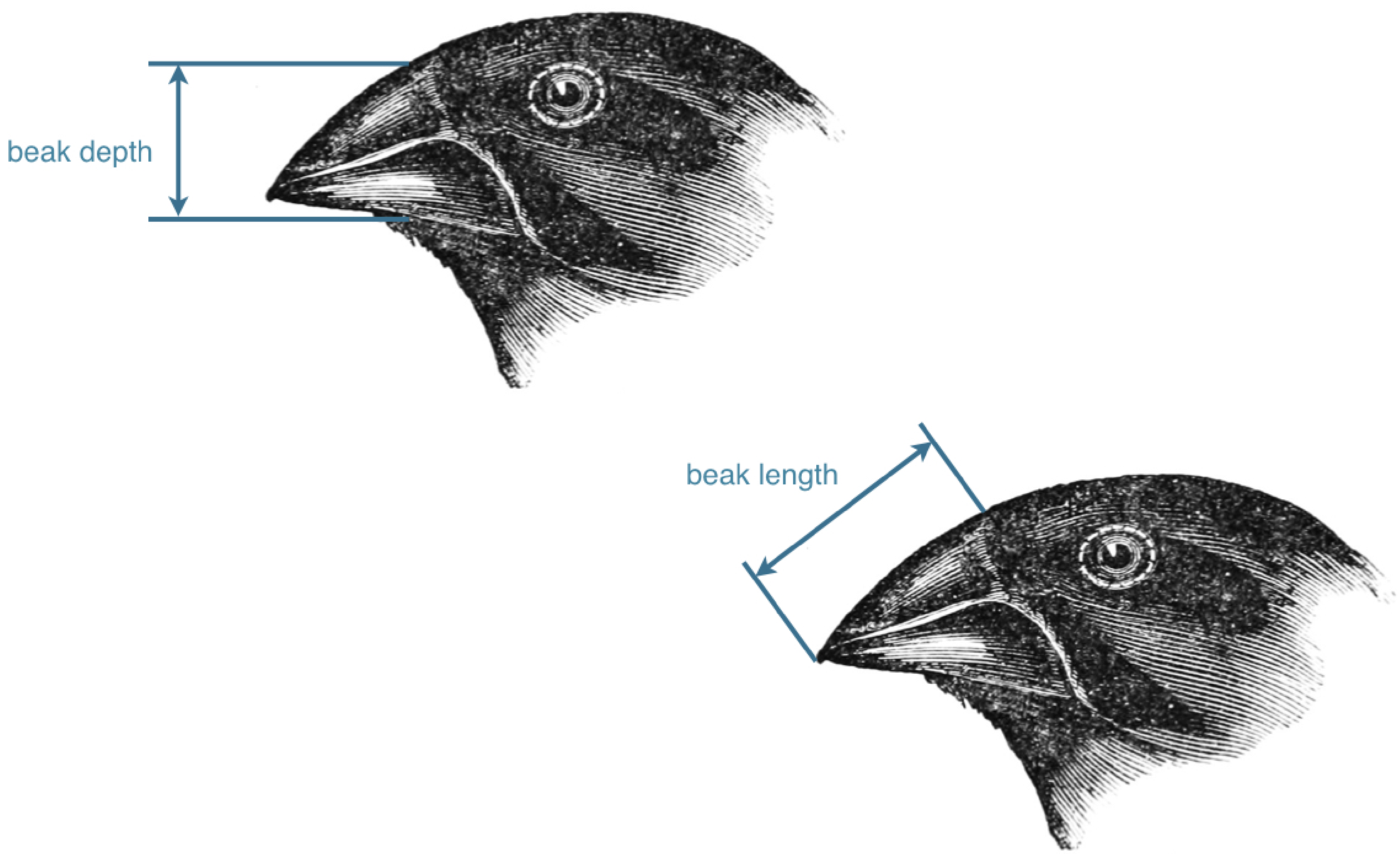
The drought of winter 1976/1977



[1] Source: Grant and Grant, 2014

Beak geometry



[1] Source: John Gould, public domain

Hint

- `draw_bs_pairs_linreg()` will come in handy

Let's practice!

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Calculation of heritability

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The finches of Daphne Major

[1] Source: John Gould, public domain

Heredity

- The tendency for parental traits to be inherited by offspring

Let's practice!

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Final thoughts

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Your statistical thinking skills

- Perform EDA
  - Generate effective plots like ECDFs
  - Compute summary statistics
- Estimate parameters
  - By optimization, including linear regression
  - Determine confidence intervals
- Formulate and test hypotheses

Let's practice!

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